10.023 Designing Energy System (DES)
1D Report Worksheet (2025)

Group: 3.8
Due Date: Week 13 Friday (15 Aug) 6 pm

Grading

This 1D is worth a total of 20% of the total marks allocated for DES based on: (1) 1D worksheet 1 (3%); (2) 1D worksheet 2 (3%); (3) product exhibit (6%); and **(4) worksheet consisting of data collection and analysis (8%)**.

Maximum 50 words Problem Statement (Include background, clear problem description, evidence, impact, and desired outcome to define and explain the issue effectively)

Desk cleaners are usually charged by **USB** which is **unsustainable** since most of Singapore's electricity comes from **burning natural gas** (Energy Market Authority, 2023). It is also **inconvenient** because people may **forget** or **lose** their **cable**. Hence, we aim to allow the desk cleaner to run on **solar energy without compromising** its **usability**.

In no more than 50 words, describe your solution clearly. Explain how it addresses the problem.

Our solution **integrates** a **solar panel** with a **desk cleaner** to increase sustainability. This allows it to be used **without being plugged in** for charging. Even with our modifications, it is still **handheld**, and **daily exposure** to the sun can provide **3 minutes** of use per day.

Explain each modification clearly (reader should be able to replicate from this instruction).

1. **Battery**
    - Attach a battery to the **charge controller**
        - Needs to have the **same connector** as the **charge controller** (we used a **pouch battery**)
            - **Detach** the **Printed Circuit Board (PCB)** from the original cylindrical battery
            - Use of the original battery is possible, but a **JST PH2.0 connector** and a **crimp** need to be used
    - The battery we used is a **3.7V, 1.2Ah**, but the actual size needed is only **3.7 V, 0.1 Ah**
2. **Solar panel**
    - Attach the solar panel to the **charge controller**
    - The solar panel we used is **6V and 1W**
    - Solar panel size is 110 mm x 60 mm, but the holder changes the exposed surface to **100 mm x 60 mm**
3. **Extend body** of the desk cleaner by **1.2 cm vertically** to house the new components
    - Insert plastic spacer between the two parts that house the electronics (3D printed)
4. Custom adjustable **solar panel holder** for ease of use
    - Allows the solar panel to be moved out of the way during use and angled at an optimal position during charging (3D printed)

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In this one page, provide details of data collection process and why the data is important.

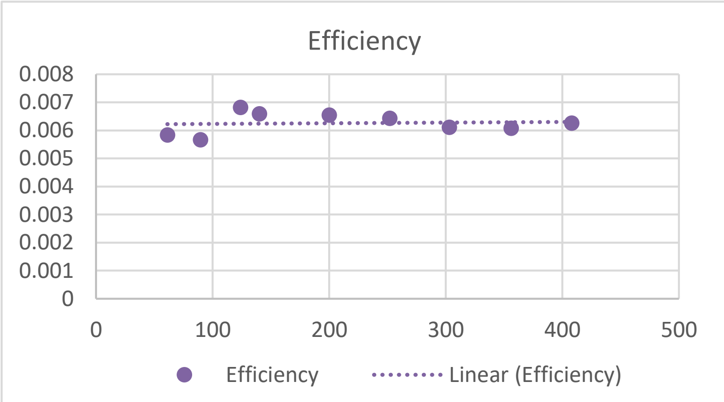
Process

1. Record the **solar irradiance** from a halogen lamp using an irradiance meter ensuring that it is measured from the same height as the solar panel.
2. Record the **voltage** and **current received** by the **battery** (after passing through the charge controller)
3. Calculate the **power** for each set of voltage and current (P = IV)
4. Repeat steps 1 to 3 for various **levels of solar irradiance** by varying the **distance** between the **lamp** and the **solar panel**
5. Plot the graph of **power** against **solar irradiance**
    - This will give us insight on how much the desk cleaner can be charged in a day

Importance

- Collecting the data allows us to confirm if the **energy provided by solar power** is **sufficient** to power the desk cleaner
    - Assuming the **daily use** of the desk is **3 minutes**, the desk cleaner needs to be **charged enough** in a **day** to function for at least that long

Irradiance (W/m^2)	V (V)	I (A)	P = IV (W)	Efficiency
408	4.642	0.033	0.153186	0.006258
356	4.636	0.028	0.129808	0.006077
303	4.632	0.024	0.111168	0.006115
252	4.626	0.021	0.097146	0.006425
200	4.622	0.017	0.078574	0.006548
140	4.612	0.012	0.055344	0.006589
124	4.611	0.011	0.050721	0.006817
89.6	4.348	0.007	0.030436	0.005661
61.2	4.286	0.005	0.02143	0.005836



Size = 0.006 m^2 (adjusted for exposed area when in the holder)

Since efficiency is **relatively constant** across different levels of irradiance, let us assume that **efficiency = 0.005 (underestimation)**
Let PSH = 3.936 h (NASA Power, n.d.) (December 2022)
Energy charged/per day = energy in x efficiency = 3.936 h x 1000 W/m^2 x 0.006 m^2 x 0.005 = 425 J (3sf)
Power consumed = 2 W (check below for power data)
Usage time/day = 425/2 = 220 s = 3.5 min (2sf)

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In this **one** page, detail the energy and power analysis of the device before and after modification. Explain the results.

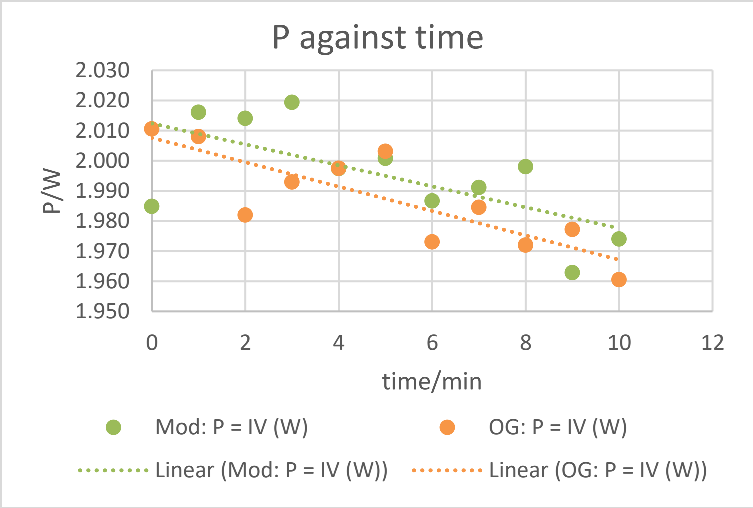
Process

1. Record the **voltage** and **current** provided by the **unmodified desk cleaner** when **first turned on** and for **every minute** for **10 minutes**
2. Calculate the **power** for each set of voltage and current (P = IV)
3. Repeat step 1 and 2 for the **modified desk cleaner**
4. Plot the graphs of **power against time** for both desk cleaners

Importance

- **Comparing** the **power** for the **unmodified desk cleaner** vs the **modified** one informs us whether our modification has **significantly changed the power output** and hence the **performance** of the device
    - If the power output **significantly decreased**, this means that our modifications might be **compromising too much** of the desk cleaner's performance
    - If there are **no significant changes**, this indicates that the goal of **maintaining functionality** while **increasing sustainability** has been achieved

	Modified			Original		
time/min	V (V)	I (A)	Mod: P = IV (W)	V (V)	I (A)	OG: P = IV (W)
0	3.662	0.542	1.985	3.649	0.551	2.011
1	3.659	0.551	2.016	3.651	0.550	2.008
2	3.642	0.553	2.014	3.650	0.543	1.982
3	3.632	0.556	2.019	3.650	0.546	1.993
4	3.625	0.551	1.997	3.645	0.548	1.997
5	3.618	0.553	2.001	3.642	0.550	2.003
6	3.612	0.550	1.987	3.647	0.541	1.973
7	3.607	0.552	1.991	3.648	0.544	1.985
8	3.600	0.555	1.998	3.645	0.541	1.972
9	3.595	0.546	1.963	3.648	0.542	1.977
10	3.589	0.550	1.974	3.644	0.538	1.960



Average power (modified) = 1.995 W
Average power (original) = 1.987 W
% Difference = (modified – original)/original x 100 % = (1.995 – 1.987)/1.987 x 100 % = 0.403 %
Especially considering a use time of 3 min per day, there is no significant difference in the performance for the modified and original desk cleaner.

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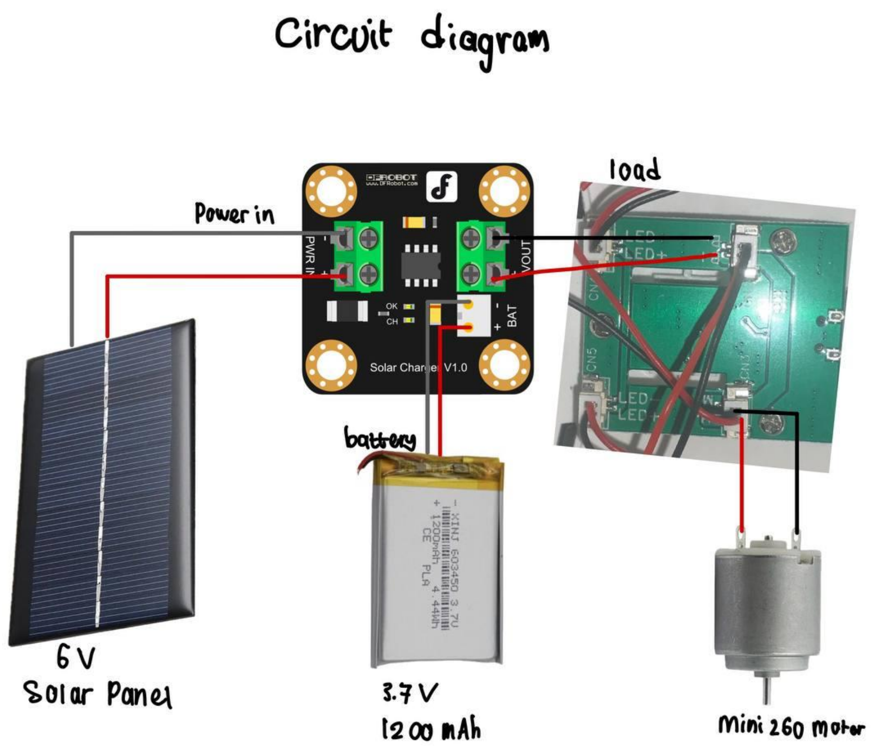
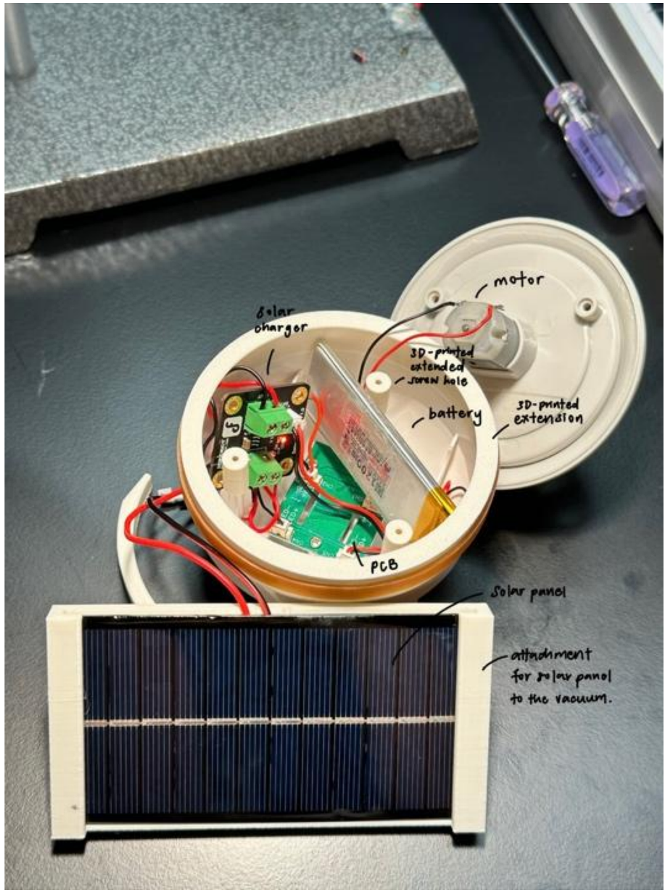
In no more than 50 words, discuss the impact on functionality and usability of the proposed solution.

There is **no significant difference** between the **power output** of the modified and unmodified desk cleaner, hence its **main purpose of cleaning** is still fulfilled. The attachments and solar panel are also **not** extremely **bulky**, which allows the desk cleaner to be used normally.

Detailed schematic of the working prototype (mechanical and/or electrical).

Electrical schematic of working prototype:





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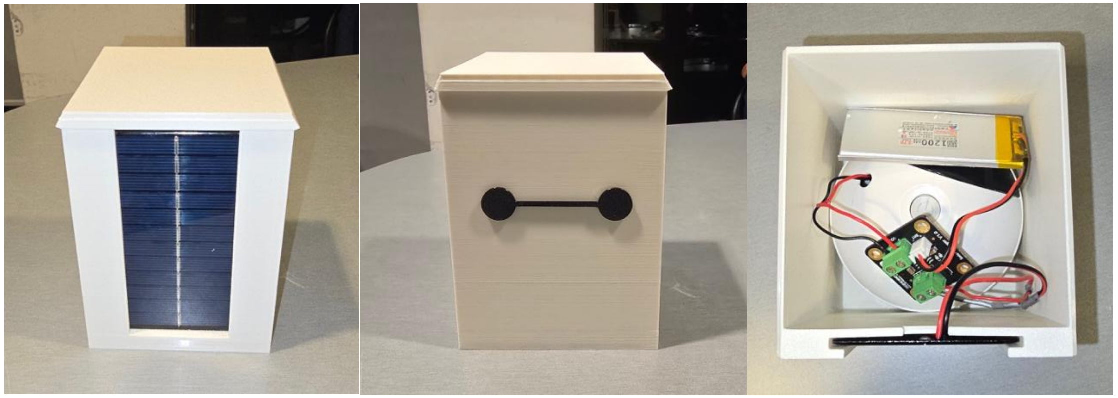
Mechanical schematic of working prototype:

We added an attachment that acts like a clip onto the side of the vacuum.
The solar panel holder is connected to the clip by a hinge attachment.
The hinge allows the panel to be kept vertically when the vacuum is in use and then turned upwards towards sunlight when charging.

Photos of working prototype

First iteration:

We realized housing the solar panel with the circuitry and vacuum in a **box** decreased the **usability** of the hand-held vacuum. It was too bulky to hold easily and the ergonomics were compromised.



Final iteration:

We improved the design by modifying the vacuum instead of simply placing it in a case for the first iteration.

The solar panel is now an added component to the vacuum and the **circuitry is fitted inside** the vacuum by **increasing the height** the vacuum using a 3D printed component.

References:

Energy Market Authority. (2023). What fuel is used to generate electricity in Singapore?
https://www.ema.gov.sg/resources/faqs/energy-supply/energy-security/what-fuel-is-used-to-generate-electricity-in-singapore

NASA Power. (n.d.). Data access viewer.
https://power.larc.nasa.gov/data-access-viewer/

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Appendix:

Battery sizing:

Nominal system Voltage = 3.7 V

Usage per day = $\frac{3}{60}$ hours/day (3 minutes)

Current needed to run motor = 0.54 A

total daily load = $\frac{3}{60} \times 0.54 = 0.027$ Ah/day

Unadjusted battery capacity = 1 day $\times$ 0.027 Ah/day = 0.027

MDOD = 80%

Capacity of MDOD = $\frac{0.027}{0.8} = 0.03375$ Ah

MDDOD = 50%

Capacity of MDDOD = $\frac{0.027}{0.5} = 0.054$ Ah

EOL = 80%

Capacity for EOL = 0.03375

Capacity adjusted out of all = 0.054 Ah

design margin = 1.1

Capacity for design margin = $1.1 \times 0.054 = 0.0594$ Ah = 59.4 mAh

Smallest practical capacity = 100 mAh

Figure 1: Battery sizing

Figure 2: Solar panel sizing

Solar panel sizing :

Nominal $V = 6\ V$

Rated power $= 1\ W$

Rated current $= \frac{1}{6} = 0.167\ A$

$AL = 1.2$

daily photovoltaic ampere hours needed = total daily load × A:L

$= 0.027\ Ah/day \times 1.2$

$= 0.0324\ Ah$

peak sun hour $= 3.936$ (Dec 2022)

standalone PV efficiency $= 70\%$

daily photovoltaic ampere-hours per chosen solar panel $= 0.7 \times 3.936h \times 0.167\ A$

$\approx 0.461\ Ah$

numbers of panels required $= \dfrac{\text{daily photovoltaic ampere hours}}{\text{daily photovoltaic ampere hours per chosen Solar panel}} = \dfrac{0.0324\ Ah}{0.461\ Ah} \approx 0.0703$

numbers of panels required = 1 panel